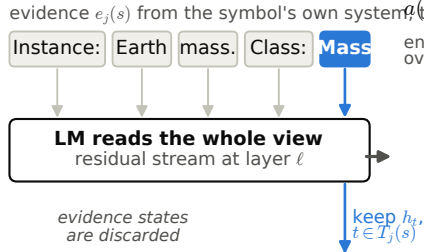


1 Native view



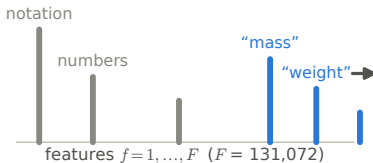
A program, a formal statement or a table column is its own single view: every token is pooled.

Ontology / KG terms: one view per instance, axiom, definition or parent (up to 16).

2 Sparse code

$$a(s) = \frac{1}{|T_j(s)|} \sum_{t \in T_j(s)} E(h_t) \in \mathbb{R}^F$$

encode each token, then average over tokens and views



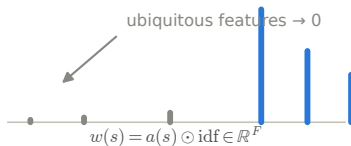
grey: features that fire on almost every symbol (notation, numbers, licence text)

blue: rare features specific to this symbol

3 Rarity weighting and matching

$$w_f(s) = a_f(s) \cdot \text{idf}_f, \quad \text{idf}_f = \log \frac{N}{1 + \text{df}(f)}$$

$\text{df}(f)$: symbols in the reference collection on which f fires



$$\hat{t}(s) = \arg \max_{t \in C(s)} \cos(w(s), w(t))$$

the cosine is a sum over shared features, so every match reads as a list of named features